AGNI PYTHARA

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Ioannina, Greece
Portfolio

Research Profile

Early-career computational photonics researcher with academic training in physics and advanced materials and research experience in nonlinear and ultrafast optics, graphene carrier dynamics, and photonic-device modelling. Experienced in physics-based numerical modelling, large-scale scientific computing, and simulation–experiment comparison, with research focused on nonlinear light–matter interaction and graphene-based photonic structures.

Research and Technical Experience

MSc Thesis Research

2025–2026 **Graphene Saturable Absorption and Optical Pulse Shaping**
University of Ioannina

- Adapted a Fortran nonequilibrium carrier-dynamics model from picosecond to femtosecond excitation and coupled it to the time-dependent optical response of multilayer photonic structures, using fourth-order Runge–Kutta (RK4) integration to investigate ultrafast nonlinear light–matter interaction.
- Developed a Python workflow to automate, manage, post-process, and visualize more than 30,000 simulations executed on a university computing cluster, enabling systematic exploration of multidimensional parameter spaces.
- Analyzed nonlinear-transmission measurements acquired by collaborators and compared them with simulations to evaluate model performance, identify physical limitations, and assess trends reported in graphene saturable-absorption literature.
- Designed and optimized asymmetric graphene-loaded Fabry–Pérot cavities through parameter sweeps, grid searches, and analytical optimization, varying material composition, cavity architecture, operating wavelength, graphene-layer number, and metallic back-mirror thickness; predicted graphene absorption enhancement from 2.3% to approximately 100%.
- Numerically demonstrated passive all-optical pulse shaping, transforming a single-peaked picosecond input into reflected profiles with two or three distinct peaks; the optimized multilayer design is progressing toward experimental realization.
- Implemented a transfer-matrix model for graphene third-harmonic generation and developed an independent finite-element COMSOL Multiphysics model for benchmarking, obtaining agreement within approximately 10%.

BSc Thesis Research

2023–2024 **Waveguide-Integrated Graphene Saturable Absorbers**
University of Ioannina

- Adapted a free-space graphene model to integrated photonic waveguides using modal properties calculated with Lumerical MODE solver.
- Designed and optimized a CMOS-compatible graphene-loaded waveguide cavity.
- Performed Fortran-based multidimensional parameter studies to investigate nonlinear absorption and critical-coupling behavior.

Education

2024–2026 **M.Sc. in Technologies of Advanced Materials**
University of Ioannina, Department of Materials Science and Engineering
Specialization in Physics and Computational Science of Advanced Materials
Thesis: *Study of graphene saturable absorption for near-infrared optical pulse reshaping*
Thesis grade: **10/10**
Course average: **8.92/10**

2018–2024 **B.Sc. in Physics**
University of Ioannina, Department of Physics
Thesis: *Graphene-Based Saturable Absorbers in Waveguide-Integrated Schemes*
Thesis grade: **10/10**

Teaching Experience

2025–2026 **Teaching Assistant**
University of Ioannina, Department of Materials Science and Engineering

- Led weekly laboratory sessions for fourth-year undergraduate students.
- Supported experiments on photocurrent measurements.
- Evaluated laboratory performance and provided technical feedback.

Conference Presentation

2024 **A. Pythara**, V. Nousia, S. Doukas and E. Lidorikis
Graphene-Based Saturable Absorbers in Waveguide-Integrated Schemes
Poster presentation
XXXVIII Panhellenic Conference on Solid State Physics and Materials Science, Ioannina, Greece

Technical Skills

Programming	Python, Fortran, C (university coursework), LaTeX
Scientific Computing	Automated simulation workflows, large-scale parameter studies, cluster computing, numerical data analysis, scientific visualization
Numerical Methods	Fourth-order Runge–Kutta (RK4), Transfer-matrix method, numerical integration
Software	COMSOL Multiphysics, Lumerical MODE, Lumerical FDTD
Research Methods	Physics-based model adaptation, literature analysis, simulation–experiment comparison, cross-method numerical benchmarking, computational device optimization, model validation and analysis of physical limitations

Languages

Greek — Native
English — C2